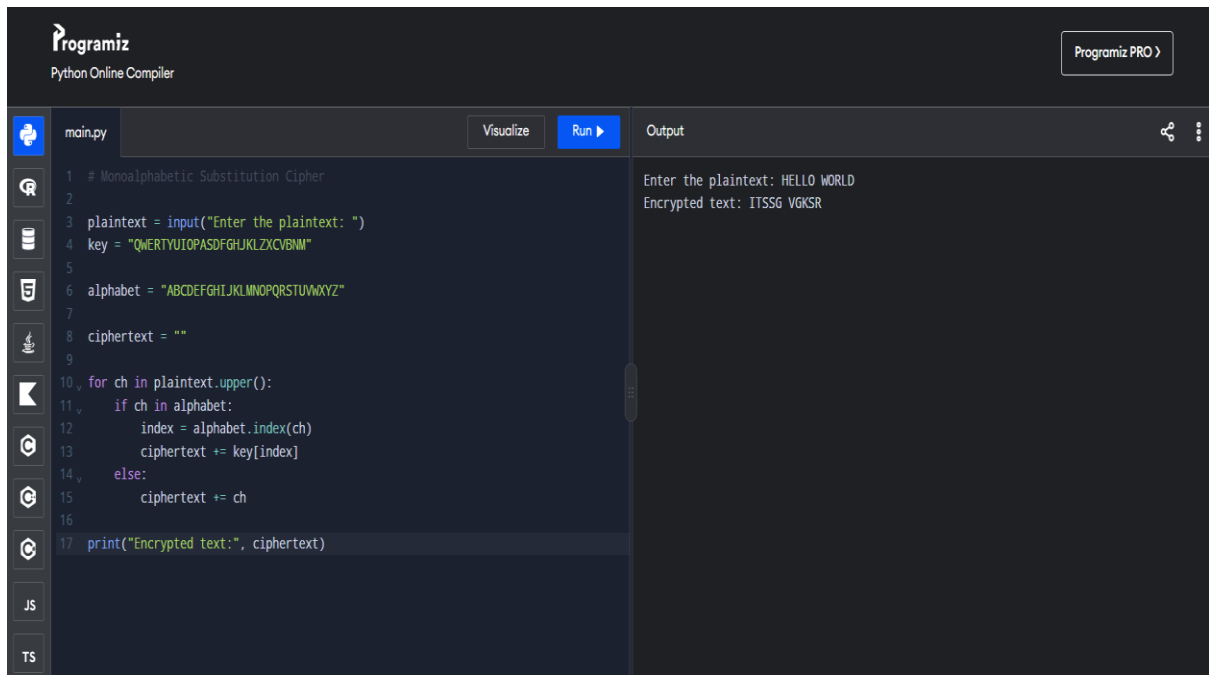


EXPERIMENT-2

2. Write a C program for monoalphabetic substitution cipher maps a plaintext alphabet to a ciphertext alphabet, so that each letter of the plaintext alphabet maps to a single unique letter of the ciphertext alphabet.



The screenshot shows the Programiz Python Online Compiler interface. The main editor displays a Python script for a monoalphabetic substitution cipher. The script prompts the user to enter a plaintext, which is then encrypted using a predefined key and alphabet. The output shows the encrypted text for the input "HELLO WORLD".

```
1 # Monoalphabetic Substitution Cipher
2
3 plaintext = input("Enter the plaintext: ")
4 key = "QWERTYUIOPASDFGHJKLZXCVBNM"
5
6 alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
7
8 ciphertext = ""
9
10 for ch in plaintext.upper():
11     if ch in alphabet:
12         index = alphabet.index(ch)
13         ciphertext += key[index]
14     else:
15         ciphertext += ch
16
17 print("Encrypted text:", ciphertext)
```

Output:

```
Enter the plaintext: HELLO WORLD
Encrypted text: ITSSG VGKSR
```